

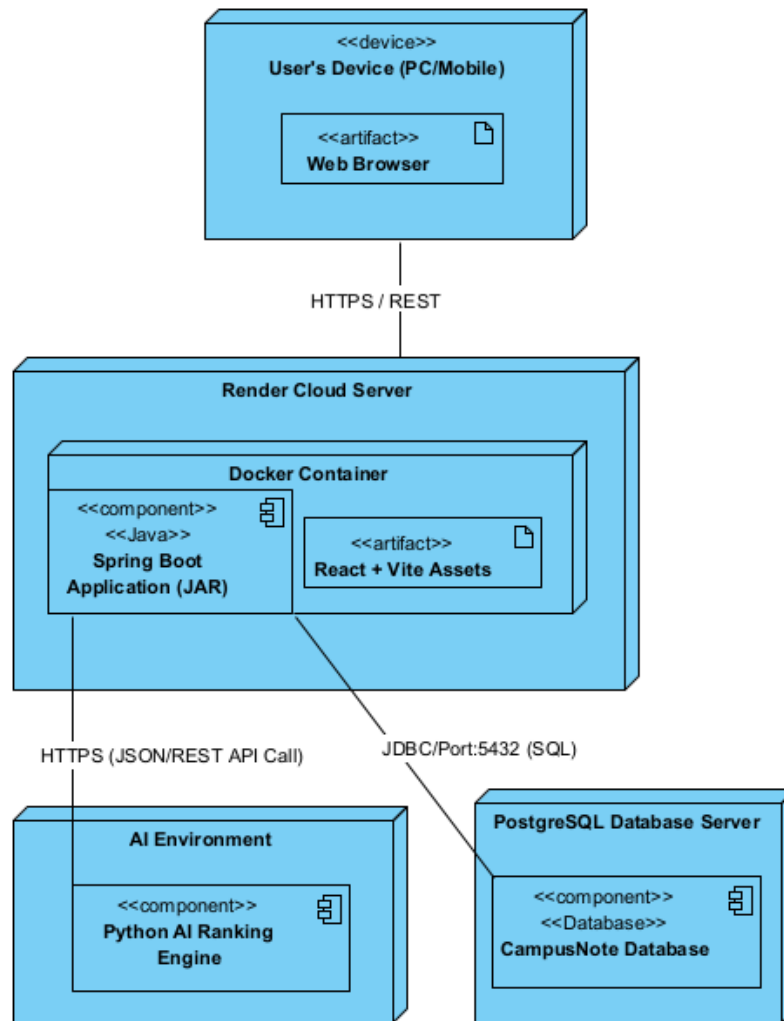


Figure 1. Deployment Diagram

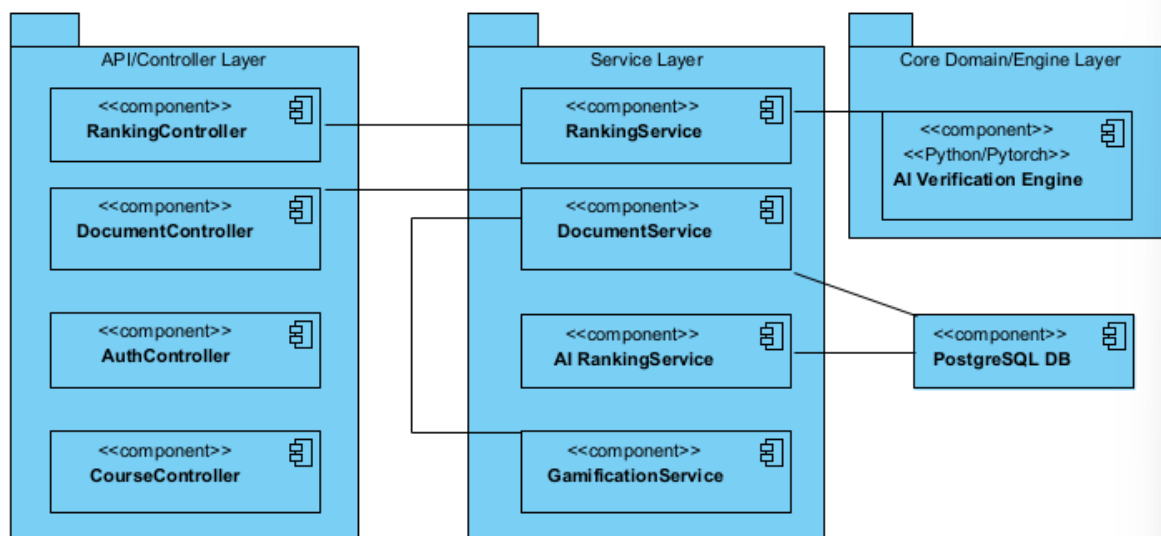


Figure 2. Component Diagram

Technology Justification

- **Frontend:** React + Vite with Tailwind CSS for high-performance, consistent UI.
- **Backend:** Java Spring Boot for secure, robust business logic.
- **AI Engine:** Python + PyTorch for specialized deep learning and PDF quality analysis.
- **Database:** PostgreSQL for ACID-compliant virtual currency (CampusCoin) integrity.
- **DevOps:** Docker and Heroku for environmental parity and seamless cloud scaling.